

Name : **Tejas Relekar** Age/Gender : 29 y / M Collected : 03/07/2024 22:34
Patient ID : **0010383069** Visit Type : IP Received : 03/07/2024 23:52
Referred By : Dr. Tushar Parmar DOB : 29/11/1994 Reported : 04/07/2024 09:07
Accession : 2400143313 Location : 1TGER`1TRGEMS4`1TBGEMS4`1EMS

DEPARTMENT OF LABORATORY MEDICINE - BIOCHEMISTRY

	Test Name	Unit	Reference Range	Low	Normal	High
	Renal Profile					
	BUN (Serum,Urease/Glutamate dehydrogenase)	mg/dL	6-20	5.89		
 MC-3469	Urea (Serum,Urease/Glutamate dehydrogenase)	mg/dL	16.6-48.5	12.60		
	Creatinine Clearance Test [eGFR]					
 MC-3469	Creatinine (Serum,Jaffe - kinetic)	mg/dL	0.70-1.20		0.88	
	eGFR (Serum,Calculated)	mL/min/1.73 m ²	Normal or High: >= 90 Mild Decreased: 60-89 Mild to Moderately Decreased: 45-59 Moderately to Severely Decreased: 30-44 Severely Decreased: 15-29 Kidney Failure: < 15		Above 90	
 MC-3469	Uric Acid (Serum,Enzymatic - colorimetric)	mg/dL	3.4-7.0		6.89	
 MC-3469	Calcium (Serum,NM - BAPTA)	mg/dL	8.6-10.0		9.68	
 MC-3469	Phosphorus (Serum,Molybdate - UV)	mg/dL	2.5-4.5		3.28	

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	Test Name	Unit	Reference Range	Low	Normal	High
	Proteins					
	Total Protein (Serum, Colorimetric - biuret)	g/dL	6.4-8.3		7.60	
	Albumin (Serum, Colorimetric - BCG)	g/dL	3.5-5.2		4.77	
	Globulin (Serum, Calculated)	g/dL	2-3.5		2.83	
	A / G ratio (Serum, Calculated)		1.0-2.5		1.69	
	S. Electrolytes (Serum, Ion-selective electrode)					
	Sodium	mmol/L	136-145		138.00	
	Potassium	mmol/L	3.5-5.1		4.33	
	Chloride	mmol/L	98-107		101.60	
	Bicarbonate (Serum, Enzymatic - PEPC)	mmol/L	22-29	21.00		

Urea:

Urea is the final degradation product of protein and amino acid metabolism. Blood urea nitrogen currently is the most widely used screening test for the evaluation of kidney function. Protein catabolism causes formation of ammonia which is synthesized to urea with and elimination of excess nitrogen in the liver. Increased blood urea nitrogen may be due to cardiac decompensation, water depletion due to decreased intake and excessive loss, increased protein catabolism, high protein diet, acute glomerulonephritis, chronic nephritis, polycystic kidney disease, nephrosclerosis, tubular necrosis and all types of

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obstruction of the urinary tract.

Creatinine:

Creatinine is synthesized endogenously from creatine and creatine phosphate and is removed from plasma by glomerular filtration into the urine. Creatinine levels in renal disease generally do not increase until renal function is substantially impaired.

Uric Acid:

- Uric acid measurements are used in the diagnosis and treatment of numerous renal and metabolic disorders, including renal failure, gout, leukemia, psoriasis, starvation or other wasting conditions, and of patients receiving cytotoxic drugs.

Calcium:

- Serum calcium levels are controlled by parathyroid hormone (PTH), calcitonin, and vitamin D.
- An imbalance in any of these modulators leads to alterations of the body and serum calcium levels.
- Increases in serum PTH or vitamin D are usually associated with hypercalcemia.
- Increased serum calcium levels may also be observed in multiple myeloma and other neoplastic diseases.
- Hypocalcemia may be observed e.g. in hypoparathyroidism, nephrosis, and pancreatitis.

Phosphorus:

- Hypoparathyroidism, vitamin D intoxication and renal failure with decreased glomerular phosphate filtration give rise to hyperphosphatemia.
- Hypophosphatemia occurs in rickets, hyperparathyroidism and Fanconi's syndrome.

Total Protein:

- Hypoproteinemia can be caused by diseases and disorders such as loss of blood, sprue, nephrotic syndrome, severe burns, salt retention syndrome and Kwashiorkor (acute protein deficiency).



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- Hyperproteinemia can be observed in cases of severe dehydration and illnesses such as multiple myeloma.

Albumin:

- Hyperalbuminemia is of little diagnostic significance except in the case of dehydration.
- Hypoalbuminemia occurs during many illnesses and is caused by several factors: compromised synthesis due either to liver disease or as a consequence of reduced protein uptake; elevated catabolism due to tissue damage (severe burns) or inflammation; malabsorption of amino acids (Crohn's disease); proteinuria as a consequence of nephrotic syndrome; protein loss via the stool (neoplastic disease).

Sodium:

- Sodium:** is the major extracellular cation and functions to maintain fluid distribution and osmotic pressure. Some causes of decreased levels of sodium include prolonged vomiting or diarrhoea, diminished reabsorption in the kidney and excessive fluid retention. Common causes of increased sodium include excessive fluid loss, high salt intake and increased kidney reabsorption.

Potassium:

- Potassium:** is the major intracellular cation and is critical to neural and muscle cell activity. Some causes of decreased potassium levels include reduced intake of dietary potassium or excessive loss of potassium from the body due to diarrhea, prolonged vomiting or increased renal excretion. Increased potassium levels may be caused by dehydration or shock, severe burns, diabetic ketoacidosis, and retention of potassium by the kidney.

Chloride:

- Chloride is the major extracellular anion and serves to regulate the balance of extracellular fluid distribution. Similarly to the other ions, common causes of decreased chloride include reduced dietary intake, prolonged vomiting and reduced renal reabsorption as well as some forms of acidosis and alkalosis. Increased chloride values are found in dehydration, kidney failure, some forms of acidosis, high dietary or parenteral chloride intake, and salicylate poisoning.



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Bicarbonate:

Bicarbonate is a significant indicator of electrolyte dispersion and anion deficit. Together with pH determination, it is used in the diagnosis and treatment of disorders associated with acid-base imbalance in the respiratory and metabolic systems like diarrhea, renal tubular acidosis, carbonic anhydrase inhibitors, hyperkalemic acidosis, renal failure and ketoacidosis.

∞ End of Report ∞


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